1. The mean and variance of a set of 15 numbers are 12 and 14 respectively. The mean and variance of another set of 15 numbers are 14 and σ^2 respectively. If the variance of all the 30 numbers in the two sets is 13, then σ^2 is equal to

[2023 (06 Apr Shift 1)]

(1) 10

(2) 11

(3) 9

(4) 12

2. If the mean and variance of the frequency distribution

x_i	2	4	6	8	10	12	14	16
f_i	4	4	α	15	8	β	4	5

are 9 and 15.08 respectively, then the value of $\alpha^2+\beta^2-\alpha\beta$ is _____.

[2023 (06 Apr Shift 2)]

3. Let the mean and variance of 8 numbers $x,\ y,\ 10,\ 12,\ 6, 12,\ 4,\ 8$ be 9 and 9.25 respectively. If $x>y$, then $3x-2y$ is equal to _______

[2023 (08 Apr Shift 1)]

4. Let the mean and variance of 12 observations be $\frac{9}{2}$ and 4 respectively. Later on, it was observed that two observations were considered as 9 and 10 instead of 7 and 14 respectively. If the correct variance is $\frac{m}{n}$, where m and n are coprime, then $m+n$ are coprime, then $m+n$ is equal to

[2023 (08 Apr Shift 2)]

(1) 315

(2) 316

(3) 314

(4) 317

5. If the mean of the frequency distribution

Class :	$0-10$	$10-20$	$20-30$	$30-40$	$40-50$
Frequency :	2	3	x	5	4

is 28, then its variance is ________ .

[2023 (10 Apr Shift 1)]

6. Let μ be the mean and σ be the standard deviation of the distribution

X_i	0	1	2	3	4	5
f_i	$k+2$	$2k$	k^2-1	k^2-1	k^2+1	$k-3$

where $\Sigma f_i=62$. If $[x]$ denotes the greatest integer $\le x$, then $[\mu^2+\sigma^2]$ is equal to

[2023 (10 Apr Shift 2)]

(1) 9

(2) 8

(3) 7

(4) 6

7. Let sets A and B have 5 elements each. Let the mean of the elements in sets A and B be 5 and 8 respectively and the variance of the elements in sets A and B be 12 and 20 respectively. A new set C of 10 elements is formed by subtracting 3 from each element of A and adding 2 to each element of B. Then the sum of the mean and variance of the elements of C is

[2023 (11 Apr Shift 1)]

(1) 40

(2) 32

(3) 38

(4) 36

8. Let the mean of 6 observations $1, 2, 4, 5, x$ and y be 5 and their variance be 10. Then their mean deviation about the mean is equal to

[2023 (11 Apr Shift 2)]

(1) $\frac{7}{3}$

(2) 3

(3) $\frac{8}{3}$

(4) $\frac{10}{3}$

9. Let the positive numbers $a_1,\ a_2,\ a_3,\ a_4$ and a_5 be in a G.P. Let their mean and variance be $\frac{31}{10}$ and $\frac{m}{n}$ respectively, where m and n are co-prime. If the mean of their reciprocals is $\frac{31}{10}$ and $a_3+a_4+a_5=14$, then $m+n$ is equal to ____________.

[2023 (12 Apr Shift 1)]

10. Let the mean of the data

x	1	3	5	7	9
Frequency (f)	4	24	28	α	8

be 5. If m and σ^2 are respectively the mean deviation about the mean and the variance of the data, then $\frac{3\alpha}{m+\sigma^2}$ is equal to _______.

[2023 (13 Apr Shift 1)]

11. The mean and standard deviation of the marks of 10 students were found to be 50 and 12 respectively. Later, it was observed that two marks 20 and 25 were wrongly read as 45 and 50 respectively. Then the correct variance is

[2023 (13 Apr Shift 2)]

12. The mean and standard deviation of 10 observations are 20 and 8 respectively. Later on, it was observed that one observation was recorded as 50 instead of 40. Then the correct variance is

[2023 (15 Apr Shift 1)]

(1) 11

(2) 13

(3) 12

(4) 14

ANSWER KEYS

1. (1)	**2.** (25)	**3.** (25)	**4.** (4)	**5.** (151)	**6.** (2)	**7.** (3)	**8.** (3)
9. (211)	**10.** (8)	**11.** (269)	**12.** (2)				

1. (1)

Given that,

For set 1 :

Mean $=12$ and Variance $=14$

For set 2 :

Mean $=14$ and Variance $=\sigma^2$

Now, For set 1 we have,

$\Rightarrow \frac{\sum_{i=1}^{i=15} x_i}{15}=12$

$\Rightarrow \sum_{i=1}^{i=15} x_i=15 \times 12$

Also, $\frac{\sum_{i=1}^{i=15} x_i{}^2}{15}-\left(\frac{\sum_{i=1}^{i=15} x_i}{15}\right)^2=14$

$\Rightarrow \frac{\sum_{i=1}^{i=15} x_i{}^2}{15}-12^2=14$

$\Rightarrow \sum_{i=1}^{i=15} x_i{}^2=(14+144) \times 15$

Now for set 2 :

$\Rightarrow \frac{\sum_{i=1}^{i=15} y_i}{15}=14$

and $\Rightarrow \sum_{i=1}^{i=15} y_i=15 \times 14$

Also, $\frac{\sum_{i=1}^{i=15} y_i{}^2}{15}-\left(\frac{\sum_{i=1}^{i=15} y_i}{15}\right)^2=\sigma^2$

$\Rightarrow \frac{\sum_{i=1}^{i=15} y_i{}^2}{15}-14^2=\sigma^2$

$\Rightarrow \sum_{i=1}^{i=15} y_i{}^2=(\sigma^2+196) \times 15$

Now,

The combined variance will be

$\Rightarrow 13=\frac{\left(\sum_{i=1}^{i=15} x_i{}^2\right) \times 15+\left(\sum_{i=1}^{i=15} y_i{}^2\right) \times 15}{15+15}-\left(\frac{\sum_{i=1}^{i=15} x_i+\sum_{i=1}^{i=15} y_i}{15+15}\right)^2$

$\Rightarrow 13=\frac{(14+144) \times 15+(\sigma^2+196) \times 15}{30}-13^2$

$\therefore \sigma^2=10$

Hence, $\sigma^2=10$

2. **(25)**

Given,

The mean and variance of the frequency distribution

x_i	2	4	6	8	10	12	14	16
f_i	4	4	α	15	8	β	4	5

are 9 and 15.08 respectively,

Now mean is given by,

Mean $= \frac{8+16+120+80+56+80+6\alpha+12\beta}{40+\alpha+\beta} = 9$

$\Rightarrow 360 + 9\alpha + 9\beta = 360 + 6\alpha + 12\beta$

$\Rightarrow 3\alpha - 3\beta = 0$

$\Rightarrow \alpha = \beta \ldots\ldots(1)$

Now variance is given by,

$15.08 = \frac{16+64+36\alpha+960+800+144\alpha+784+1280}{40+2\alpha} - 9^2$

$\Rightarrow 96.08 = \frac{3904+180\alpha}{40+2\alpha}$

$\Rightarrow (40 + 2\alpha)(96.08) = 3904 + 180\alpha$

$\Rightarrow 3843.20 + (192.16)\alpha = 3904 + 180\alpha$

$\Rightarrow (12.16)\alpha = 60.80$

$\Rightarrow \alpha = 5 = \beta$

Hence, $\alpha^2 + \beta^2 - \alpha\beta = 25 + 25 - 25 = 25$

3. **(25)**

Given, the data $x,\ y,\ 10,\ 12,\ 4,\ 6,\ 8,\ 12$.

So,

Mean $= \frac{x+y+10+12+4+6+8+12}{8}$

$\Rightarrow 9 = \frac{x+y+52}{8}$

$\Rightarrow x + y + 52 = 72$

$\Rightarrow x + y = 20$

And,

Variance $= \left(\frac{\sum x_i^2}{n}\right) - \left(\frac{\sum x_i}{n}\right)^2$

$\Rightarrow 9.25 = \left(\frac{x^2+y^2+100+144+16+36+64+144}{8}\right) - (9)^2$

$\Rightarrow x^2 + y^2 = 218$

$\Rightarrow (x + y)^2 - 2xy = 218$

$\Rightarrow 20^2 - 2xy = 218$

$\Rightarrow 2xy = 182$

$\Rightarrow xy = 91$

Now solving $x + y = 20$ & $xy = 91$ we get,

$x = 13,\ y = 7 \Rightarrow 3x - 2y = 25$

4. (4)

Given,

The mean and variance of 12 observations be $\frac{9}{2}$ and 4 respectively,

So, mean will be,

$\bar{x}=\frac{x_1+x_2+\ldots\ldots+9+10+\ldots\ldots+x_{12}}{12}=\frac{9}{2}$

$\Rightarrow x_1+x_2\ldots\ldots+x_{12}+19=54$

Now removing the observation 9 & 10 and adding the observation 7 & 14,

So, the new total sum of the observation will be,

$x_1+x_2+\ldots\ldots+x_{12}+7+14=54-19+7+14$

Now new mean will be,

$\frac{x_1+x_2+\ldots+x_{12}+7+14}{12}=\frac{56}{12}$

$\Rightarrow \bar{x}_{\text{new}}=\frac{14}{3}$

Now using the formula of the variance we get,

$\frac{x_1^2+x_2^2+\ldots\ldots+x_{12}^2+9^2+10^2}{12}-\left(\frac{9}{2}\right)^2=4$

$\Rightarrow x_1^2+x_2^2+\ldots\ldots+x_{12}^2+9^2+10^2-81\times 3=4\times 12$

$\Rightarrow x_1^2+x_2^2+\ldots\ldots+x_{12}^2+9^2+10^2=291$

Now removing 9^2 & 10^2 and adding 7^2 & 14^2 we get,

$\Rightarrow x_1^2+x_2^2+\ldots\ldots+x_{12}^2+7^2+14^2=355$

New variance $=\frac{\sum x_i^2}{N}-\left(\bar{x}\right)^2$

$=\frac{355}{12}-\left(\frac{14}{3}\right)^2$

$=\frac{281}{36}$

Now on comparing with $\frac{m}{n}=\frac{281}{36}$ we get,

$\Rightarrow m=281$ and $n=36$

$\therefore\ m+n=317$

5. (151)

Given that the mean for the frequency distribution is 28.

We know that Mean $(\overline{x})=\frac{\sum f_i x_i}{\sum f_i}$

Now for the class interval $0-10$ we get $x_1=\frac{0+10}{2}=5$.

Similarly we can calculate x_i for other class intervals.

Now we know that mean $=28$

$\Rightarrow \frac{2\times 5+3\times 15+x\times 25+5\times 35+4\times 45}{14+x}=28$

$\Rightarrow x=6$

Also we know that variance $\left(\sigma^2\right)=\frac{\sum f_i\left(\bar{x}-x_i\right)^2}{\sum f_i}$

$\sigma^2=\frac{2\times(28-5)^2+3\times(28-15)^2+6\times(28-25)^2+5\times(28-35)^2+4\times(28-45)^2}{20}$

$\sigma^2=\frac{2\times(23)^2+3\times(13)^2+6\times(3)^2+5\times(7)^2+4\times(17)^2}{20}$

$\sigma^2=151$

Hence this is the required answer.

6. (2)

Given,

X_i	0	1	2	3	4	5
f_i	$k+2$	$2k$	k^2-1	k^2-1	k^2+1	$k-3$

Now we know that,

Mean $=\frac{\sum X_i f_i}{\sum f_i}$

$\Rightarrow \mu=\frac{2k+2k^2-2+3k^2-3+4k^2+4+5k-15}{62}$

$\Rightarrow \mu=\frac{9k^2+7k-16}{62}\ldots\ldots(1)$

And $\sum f_i=62$

$\Rightarrow 3k^2+4k-2=62$

$\Rightarrow 3k^2+4k-64=0$

$\Rightarrow (3k+16)(k-4)=0$

$\Rightarrow k=4$

Now putting the value of k in above equation (1) we get,

$\mu=\frac{156}{62}$

Now finding variance we get,

$\sigma^2=\frac{\sum (X_i)^2 f_i}{\sum f_i}-(\text{Mean})^2$

$\Rightarrow \sigma^2=\frac{1\cdot(2k)+4(k^2-1)+9(k^2-1)+16(k^2+1)+25(k-3)}{62}-\left(\frac{156}{62}\right)^2$

$\Rightarrow \sigma^2=\frac{29k^2+27k-72}{62}-\mu^2$

Now putting the value of k in above equation we get,

$\Rightarrow \sigma^2=\frac{500}{62}-\mu^2$

$\Rightarrow \sigma^2+\mu^2=\frac{500}{62}$

$\Rightarrow [\sigma^2+\mu^2]=8$

7. (3)

Let the elements in set A be $\{x_1,x_2,x_3,x_4,x_5\}$

Now given mean is $=5$

Now subtracting 3 from each term we get,

New mean as $(x_1-3,x_2-3\ldots,x_5-3)=5-3=2$

So, sum of elements will be $2\times 5=10$

Also given variance,

$\text{Var}(X)=12$

Now we know that subtracting 3 from each term will not change the variance,

So, $\text{Var}(x_1-3,x_2-3,\ldots x_5-3)=12$

$\Rightarrow \frac{\sum (x_i-3)^2}{5}-4=12$

$\Rightarrow \sum (x_i-3)^2=80$

Now let elements in set B be $\{y_1,y_2\ldots y_5\}$

Given mean $(y_1,y_2\ldots y_5)=8$

Now adding each element by 2 we get,

New mean $(y_1+2,y_2+2,\ldots y_5+2)=10$

So, sum of elements will be $10\times 5=50$

Also given $\text{Var}(y_1,y_2\ldots y_5)=20$

Similarly new variance,

$\text{Var}(y_1+2,y_2+2\ldots y_5+2)=20$

$\Rightarrow \frac{\sum (y_i+2)^2}{5}-100=20$

$\Rightarrow \sum (y_i+2)^2=120\times 5$

Now finding, combined mean we get,

$\frac{\sum_{i=1}^{5}(x_i-3)+\sum (y_i+2)}{10}=\frac{10+50}{10}=6$

And combined variance $=\frac{\sum (x_i-3)^2+\sum (y_i+2)^2}{10}-6^2$

$=\frac{80+120\times 5}{10}-36=32$

Hence, the sum of combined mean and variance will be $32+6=38$

8. **(3)**

Consider the data $1, 2, 4, 5, x, y$.

So,

Mean $= \frac{1+2+4+5+x+y}{6}$

$\Rightarrow 5 = \frac{x+y+12}{6}$

$\Rightarrow x + y = 18$

And,

Variance $= \left(\frac{\sum x_i^2}{n}\right) - \left(\frac{\sum x_i}{n}\right)^2$

$\Rightarrow 10 = \left(\frac{1^2+2^2+4^2+5^2+x^2+y^2}{6}\right) - (5)^2$

$\Rightarrow x^2 + y^2 = 164$

We have $x + y = 18$ and $x^2 + y^2 = 164$.

Now solving above equations we get, $x = 10,\ y = 8$,

Now mean deviation about mean will be $= \frac{\sum_{i=1}^{6} |x_i - \overline{x}|}{6}$

$= \frac{|1-5|+|2-5|+|4-5|+|5-5|+|10-5|+|8-5|}{6}$

$= \frac{16}{6}$

$= \frac{8}{3}$

Therefore, the required answer is $\frac{8}{3}$

9. **(211)**

5 positive numbers $a_1,\ a_2, \ldots,\ a_5$ are in geometric progression. So, let the numbers be $\frac{a}{r^2},\ \frac{a}{r},\ a,\ ar,\ ar^2$

According to question,

$\frac{\frac{a}{r^2}+\frac{a}{r}+a+ar+ar^2}{5} = \frac{31}{10} \quad \ldots(1)$

And,

$\frac{\frac{r^2}{a}+\frac{r}{a}+\frac{1}{a}+\frac{1}{ar}+\frac{1}{ar^2}}{5} = \frac{31}{40}$

$\Rightarrow \frac{\frac{1}{a}\left(r^2+r+1+\frac{1}{r}+\frac{1}{r^2}\right)}{5} = \frac{31}{40} \quad \ldots(2)$

By (1) & (2), we get

$a^2 = 4\ \Rightarrow a = 2$

So,

$\left(r^2 + r + 1 + \frac{1}{r} + \frac{1}{r^2}\right) = \frac{31}{4}$

$\Rightarrow \left(r + \frac{1}{r}\right)^2 + r + \frac{1}{r} = \frac{27}{4} + 2$

$\Rightarrow t^2 + t = \frac{35}{4}$

where, $t = r + \frac{1}{r}$

$4t^2 + 4t - 35 = 0$

$\Rightarrow t = \frac{5}{2} \Rightarrow r = 2$

So, numbers are $\frac{1}{2},\ 1,\ 2,\ 4,\ 8$

Variance is

$= \frac{\frac{1}{4}+1+4+16+64}{5} - \left(\frac{\frac{1}{2}+1+2+4+8}{5}\right)^2$

$= \frac{186}{25} = \frac{m}{n}$

$m + n = 211$

Hence this is the required option.

10. (8)

Given,

The mean of the below data is 5

x	1	3	5	7	9
Frequency (f)	4	24	28	α	8

Now finding mean we get,

$\frac{4+72+28\times5+7\alpha+72}{64+\alpha}=5$

$\Rightarrow\alpha=16$

So, total number of observation will be, $\sum f_i=80$

Now finding, mean deviation we get,

$\text{M.D}=\frac{\sum f_i\left|x_i-5\right|}{\sum f_i}$

$=\frac{4\times4+24\times2+0+16\times2+8\times4}{80}=\frac{8}{5}$

So, $m=\frac{8}{5}$

Now finding, variance we get,

$\sigma^2=\frac{\sum f_i\left|x_i-5\right|^2}{\sum f_i}$

$\Rightarrow\sigma^2=\frac{4\times16+24\times4+0+16\times4+8\times16}{80}$

$\Rightarrow\sigma^2=\frac{22}{5}$

Hence, the value of $\frac{3\alpha}{m+\sigma^2}=\frac{3\times16}{\frac{8}{5}+\frac{22}{5}}=8$

11. (269)

Let the observations be $x_1,\ x_2,\ x_3,\ \ldots,\ x_8,\ 45,\ 50,$

Mean$=50$

$\Rightarrow\frac{x_1+x_2+x_3+\ldots+x_8+45+50}{10}=50$

$\Rightarrow x_1+x_2+x_3+\ldots+x_8=405\ \ldots(\text{i})$

Hence, new mean

$(\bar{X})_{\text{new}}=\frac{405+20+25}{10}=45$

Now,

$\text{S.D}=\sqrt{\frac{\sum_{i=1}^{8}x_i^2+45^2+50^2}{10}-(50)^2}$

$\Rightarrow12=\sqrt{\frac{\sum_{i=1}^{8}x_i^2+4525}{10}-2500}$

$\Rightarrow\sum_{i=1}^{8}x_i^2=21915$

Now,

$(\text{Variance})_{\text{new}}=\frac{\sum_{i=1}^{8}x_i^2+20^2+25^2}{10}-(45)^2$

$\Rightarrow(\text{Variance})_{\text{new}}=\frac{21915+20^2+25^2}{10}-(45)^2$

$\Rightarrow(\text{Variance})_{\text{new}}=2294-2025$

$\Rightarrow(\text{Variance})_{\text{new}}=269$

Hence this is the correct answer.

12. (2)

Incorrect mean is

$$\frac{\sum_{i=1}^{10} x_i}{10} = 20$$

$\Rightarrow \sum_{i=1}^{10} x_i = 200 \quad \ldots.(i)$

So, corrected mean is

$\frac{\sum_{i=1}^{10} x_i - 50 + 40}{10} = \frac{200-10}{10} = 19$

Incorrect variance is

$\frac{\sum_{i=1}^{10} x_i^2}{10} - 400 = 64$

$\Rightarrow \sum_{i=1}^{10} x_i^2 = 4640$

Now, corrected variance is

$= \frac{\sum_{i=1}^{9} x_i^2 + 40^2 - 50^2}{10} - (19)^2$

$= \frac{4640-900}{10} - (19)^2$

$= 374 - 361$

$= 13$